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Experiment No:-3

QUICK SORT :

CODE:-

```
#include<stdio.h>
void quicksort(int a[50],int first,int last){
  int i, j, pivot, temp;
  if(first<last){
    pivot=first;
    i=first;
    j=last;
    while(i<j){
      while(a[i]<=a[pivot]&&i<last)
        i++;
      while(a[j]>a[pivot])
        j--;
      if(i<j){
        temp=a[i];
        a[i]=a[j];
        a[j]=temp;
      }
    }
```

```
      temp=a[pivot];
      a[pivot]=a[j];
      a[j]=temp;
      quicksort(a,first,j-1);
      quicksort(a,j+1,last);
   }
}

int main(){
   int i, n, a[50];
   printf("Enter no of elements: ");
   scanf("%d",&n);
   printf("Enter %d elements: ", n);
   for(i=0;i<n;i++)
      scanf("%d",&a[i]);
   quicksort(a,0,n-1);
   printf("sorted array: ");
   for(i=0;i<n;i++)
      printf(" %d",a[i]);
   return 0;
}
```

Output:

```
Enter no of elements: 4
Enter 4 elements: 44
11
88
33
sorted array:  11 33 44 88
```